

CHAGYL, Turkmenistan

WMO: 385110

Lat: 40.780N

Lon: 55.382E

Elev: 377

StdP: 14.50

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.4	9.1	-3.9	4.5	4.7	1.9	6.1	10.9	22.4	35.1	20.1	37.7	4.0	90	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.1	106.8	70.7	104.3	70.5	101.8	69.2	77.0	97.2	74.1	95.7	72.0	94.7	7.6	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.6	114.1	90.2	66.7	99.5	88.0	63.9	90.2	83.7	40.7	95.5	37.8	95.4	35.9	94.5	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.5	18.9	17.3	-1.9	111.0	10.3	2.9	-9.3	113.1	-15.3	114.8	-21.1	116.4	-28.6	118.5
			-2.4	78.7	10.0	3.2	-9.6	81.0	-15.5	82.9	-21.1	84.6	-28.3	86.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	59.8	32.0	33.4	48.4	59.8	74.5	83.9	88.8	84.8	73.3	57.8	43.4	35.0
	DBStd	21.76	10.17	11.25	7.94	8.30	5.91	5.74	5.16	6.02	6.41	8.64	9.15	8.49
	HDD50	1886	558	468	124	14	0	0	0	0	0	30	227	465
	HDD65	4460	1022	885	514	197	5	0	0	0	8	252	648	929
	CDD50	5446	1	3	75	308	760	1017	1204	1079	698	271	29	1
	CDD65	2549	0	0	1	42	301	567	739	614	257	28	0	0
	CDH74	36048	0	0	51	520	3341	8097	11585	9094	3023	329	8	0
	CDH80	21447	0	0	9	182	1571	4923	7582	5695	1400	83	2	0
Wind	WSAvg	7.2	6.3	7.4	7.5	8.4	7.8	7.8	7.2	6.8	7.6	6.5	6.3	6.8
Precipitation	PrecAvg	5.9	0.7	0.8	1.1	0.6	0.5	0.2	0.2	0.1	0.2	0.3	0.6	0.6
	PrecMax	8.6	1.9	1.8	2.4	1.4	1.4	0.7	0.6	0.4	0.9	0.9	2.0	1.5
	PrecMin	3.6	0.1	0.2	0.4	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0
	PrecStd	1.5	0.4	0.4	0.5	0.4	0.4	0.2	0.2	0.1	0.2	0.2	0.5	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.5	62.1	80.9	92.1	99.3	107.2	111.0	109.5	103.1	87.8	71.9	60.4
		MCWB	48.6	51.2	58.9	64.8	67.6	69.8	73.1	70.2	74.6	61.8	54.7	47.5
	2%	DB	51.4	55.6	73.1	84.9	95.4	104.7	108.0	105.2	95.4	81.6	65.9	53.7
		MCWB	45.0	47.4	55.9	62.1	66.0	69.3	71.2	70.6	65.2	60.3	51.8	46.4
	5%	DB	46.6	50.7	67.7	80.0	92.4	101.7	105.3	102.6	91.8	77.3	61.1	49.5
		MCWB	42.5	44.2	53.9	59.5	64.7	68.5	71.2	69.1	64.7	58.1	50.3	44.7
	10%	DB	43.3	46.4	62.4	74.9	88.7	98.8	102.5	99.9	88.0	73.3	57.2	46.2
		MCWB	40.1	41.7	51.0	56.7	63.0	67.1	70.1	68.3	63.0	55.8	48.3	42.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.5	52.3	60.8	70.6	70.6	74.5	80.8	80.5	76.8	66.9	59.0	50.2
		MCDB	57.0	60.9	75.9	86.1	92.9	97.5	98.5	101.1	95.3	81.4	67.2	56.9
	2%	WB	45.3	48.2	56.8	64.6	68.5	72.5	78.3	75.8	69.7	62.6	54.1	47.4
		MCDB	50.2	55.1	70.9	79.8	88.8	96.2	98.6	96.0	89.5	77.6	62.4	52.5
	5%	WB	42.6	44.8	54.3	61.0	66.9	70.8	75.3	73.0	66.9	60.0	51.2	45.0
		MCDB	46.3	49.6	66.7	75.2	85.6	94.6	95.6	95.5	85.7	72.7	58.6	49.2
	10%	WB	40.2	42.3	51.6	58.7	65.4	69.3	72.8	70.5	64.5	57.6	48.8	42.4
		MCDB	43.1	46.1	61.6	72.8	82.8	92.7	95.5	95.2	84.5	70.5	56.0	46.0
Mean Daily Temperature Range	5% DB	MDBR	14.5	15.8	20.5	21.0	22.5	23.7	23.1	24.9	25.1	23.0	18.9	14.4
		MCDBR	19.8	22.7	30.0	29.7	28.2	27.9	27.2	28.9	30.3	28.5	26.3	20.5
		MCWBR	14.4	15.1	15.9	14.0	11.8	11.6	11.3	12.6	14.6	15.4	16.1	14.6
	5% WB	MCDBR	19.5	21.2	28.0	26.3	24.3	24.6	23.0	25.4	27.2	24.5	23.0	18.8
		MCWBR	14.4	14.7	15.4	13.8	11.2	11.3	11.5	13.8	15.2	14.8	15.6	13.9
Clear-Sky Solar Irradiance	taub		0.400	0.439	0.492	0.564	0.548	0.540	0.549	0.513	0.485	0.472	0.424	0.398
	taud		2.027	1.928	1.839	1.716	1.792	1.820	1.809	1.875	1.906	1.916	2.008	2.032
	Ebn at Noon		223	234	237	229	238	240	236	240	236	220	212	211
	Edh at Noon		38	49	59	72	68	66	67	61	55	49	38	35
All-Sky Solar Radiation	RadAvg		566	848	1300	1664	2086	2390	2271	2092	1715	1158	753	508
	RadStd		72	116	102	162	98	76	84	93	65	81	75	69

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page